**NAME:ALAN PIOSAM A**
**REG NO:411625104004**
**DEPARTMENT:CSE**
**SUBJECT:PYTHON**
**BATCH:03**

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# Dice Game: Player vs Computer in Python

## Aim

To create a Python dice game where the player can choose to roll or hold each turn, competing against the computer to reach a target score of 50.

## Procedure

1. Import the random module to simulate dice rolls.
2. Define roll_dice() to generate a random number between 1 and 6.
3. Create player_turn() allowing the player to:
    - Roll the dice and add to the turn score.
    - Lose the turn if a 1 is rolled.
    - Hold to add the turn score to the total score.
4. Create computer_turn() with a simple AI: roll until turn score ≥ 10 or rolls a 1.
5. Initialize total scores for player and computer and set the target score to 50.
6. Loop turns alternately between player and computer until one reaches the target.
7. Display total scores after each turn and announce the winner at the end.

## Code:

```
import random

def roll_dice():
    return random.randint(1, 6)

def player_turn():
    turn_score = 0
    while True:
        choice = input("Roll or Hold? (r/h): ").lower()
```

```
        if choice == 'r':
            roll = roll_dice()
            print(f"You rolled: {roll}")

            if roll == 1:
                print("□ You lost this turn!")
                return 0
            else:
                turn_score += roll
                print(f"Turn Score: {turn_score}")

        elif choice == 'h':
            return turn_score
        else:
            print("Invalid choice!")

def computer_turn():
    turn_score = 0

    while turn_score < 10:  # simple AI strategy
        roll = roll_dice()
        print(f"Computer rolled: {roll}")

        if roll == 1:
            print("💻 Computer lost this turn!")
            return 0
        else:
            turn_score += roll

    return turn_score

def game():
    player_score = 0
    computer_score = 0
    target = 50

    print("🎯 First to reach 50 wins!\n")

    while player_score < target and computer_score < target:
        print("\n--- Your Turn ---")
        player_score += player_turn()
        print(f"Your Total Score: {player_score}")

        if player_score >= target:
            break

        print("\n--- Computer Turn ---")
        computer_score += computer_turn()
        print(f"Computer Total Score: {computer_score}")
```

```
    print("\n🏁 Game Over!")
    if player_score > computer_score:
        print("🎉 You Win!")
    else:
        print("💻 Computer Wins!")

game()
```

# Output:

```
Output
🎯 First to reach 50 wins!


--- Your Turn ---
Roll or Hold? (r/h): r
You rolled: 5
Turn Score: 5
Roll or Hold? (r/h): h
Your Total Score: 5

--- Computer Turn ---
Computer rolled: 4
Computer rolled: 1
💻 Computer lost this turn!
Computer Total Score: 0

--- Your Turn ---
Roll or Hold? (r/h):
```

# Result

The game correctly allows the player to roll or hold, simulates the computer's turn, and tracks scores until one reaches 50.

# Conclusion

The dice game effectively demonstrates decision-making, turn-based logic, and random number generation in Python, providing an interactive and strategic gameplay experience.